INSTITUT D'ENSEIGNEMENT SUPÉRIEUR DE RUHENGERI

Accredited by Ministerial Order N° 005/2010/Mineduc of 16 June 2010

FACULTY OF SCIENCES AND INFORMATION TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE OPTION OF INDUSTRIAL INFORMATION TECHNOLOGY

Embedded system based on smart helmet alcohol detection and prevention for motorcycles using an MQ-3 sensor

A CASE STUDY: MUSANZE

A dissertation submitted in partial fulfillment of the requirements for the award of a Bachelor's Degree of Science with Honors in Computer Science, option of Industrial Information Technology.

By: DUKUNDIGIHOZO Benitha

Reg N°: 23/23456

Supervisor: Dr. NIRERE Adria

Co-supervisor: Dr. NTEZIRIZA NKERABAHIZI Josbert

Musanze, July 2025

SCIENTIA ET LUX

B.P. 155
Ruhengeri
Rwanda
Tel : +(250) 07 88 90 30 30
: +(250) 08 88 90 30 32
Website : www.ines.ac.rw
Email :info@ines.ac.rw

DECLARATION OF ORIGINALITY

I do hereby declare that the work presented in this dissertation is my own contribution to the best of my knowledge. The same work has never been submitted to any other University or Institution. I, therefore declare that this work is my own for the partial fulfilment of the award of a Bachelor's degree with honors in Computer Science, option of Industrial Information Technology at INES-Ruhengeri.

Candidate's name: DUKUNDIGIHOZO Benitha

Candidate's signature: .....

Submission date: .....

APPROVAL

This is to certify that this dissertation entitled “Embedded system based on a smart helmet, alcohol detection and prevention for motorcycles using an mq-3 sensor” is an original study conducted by DUKUNDIGIHOZO Benitha under my supervision and guidance.

Supervisor’s name: Dr. NIRERE Adria

Supervisor’s signature: .....

Submission date: .....

DEDICATION


This is dedicated to:
The Heavenly Living God.
My beloved family.
My lecturer and supervisor
My friends and colleagues.

ACKNOWLEDGEMENTS

First and foremost, I would like to thank the Almighty God for granting me the strength, wisdom, and guidance throughout the course of this project. My deepest gratitude goes to my supervisor, Dr. NIRERE Adria, for her valuable support, constructive feedback, and constant encouragement during the development of this research. I sincerely appreciate the time and effort she invested in guiding me towards the successful completion of this work.

I extend my heartfelt thanks to my family for their unconditional love, patience, and continuous moral and financial support throughout my studies. I also wish to express my appreciation to my friends and colleagues who offered their encouragement and valuable insights during the entire research process.

Special thanks go to the motorcycle riders and traffic police officers in Musanze who provided essential feedback and willingly participated in the testing phase of the smart helmet system. Finally, I am grateful to INES-Ruhengeri for providing me with the necessary resources and a supportive learning environment to carry out this project.

ABSTRACT

Motorcycle accidents caused by drunk riding are a big problem, especially in places where motorcycles are the main type of transport. Many riders still drive after drinking alcohol, which leads to many road accidents and deaths. Current safety rules and public campaigns are not enough to fully solve this issue. There is a gap because most systems do not check alcohol levels in real-time or alert the right people. This project developed to fill that gap by making a smart helmet that uses an MQ-3 sensor to detect alcohol when alcohol exceed 0.80% BAC. The helmet checks the rider's breath and gives warning signals using lights, a buzzer, and an LCD screen. The helmet system also helps remind the rider of the danger of drunk riding. This can help increase awareness and reduce the risk of drunk driving accidents.

Data collection included interviews with riders and traffic police, as well as testing of a prototype helmet. Results showed that 65% of riders admitted riding after drinking, and 85% of traffic officers agreed that real-time alerts would aid enforcement. The helmet prototype detected alcohol accurately in 70% of test cases, sent alerts within 10 seconds, and supported by 70% of riders for real-world use. The findings show that a smart helmet system is both technically feasible and socially acceptable for reducing alcohol-related motorcycle accidents. This project contributes a proactive technological solution for road safety enforcement. Future improvements could include Bluetooth connectivity, machine learning for better detection, and broader testing in different regions.

Table of content

DECLARATION OF ORIGINALITY ii

APPROVAL iii

DEDICATION iv

ACKNOWLEDGEMENTS v

ABSTRACT vi

LIST OF FIGURES x

APPENDIX xi

LIST OF ABBRVIATIONS AND ACRONYMS xii

CHAPTER ONE: GENERAL INTRODUCTION 1

1.1 Background of the study 1

1.2. Problem statement 2

1.3 Research objectives 3

1.3.1 general objective 3

1.3.2 specific objectives 3

1.4. Research questions 3

1.5. Research hypothesis 3

1.6. Choice of the study 4

1.7. The significance of the study 4

1.7.1. personal interest 5

1.7.2. community interest 5

1.9. Research methodology 6

1.10 Organization of the Study 7

1.11 Summary 7

CHAPTER II: LITERATURE REVIEW 8

2.1 Introduction 8

2.2 Key concept definitions 8
2.2.1 Embedded system 8
2.2.2 Real-time monitoring 8
2.2.3 Mq-3 sensor 8
2.2.4 S mart helmet 9
2.2.5 Arduino nano 9
2.2.6 GSM module (sim800l) 10
2.2.7 GSM module neo-6m 10
2.2.8 LCD display (16x2) 11
2.2.9 Buzzer alarm 11
2.2.10 Switch on/off 12
2.3 Theoretical framework 12
2.4 Related work 13
2.5 Summary 15
CHAPTER 3: RESEARCH METHODOLOGY 16
3.1 Introduction 16
3.2 Case study 16
3.3 Research design 17
3.4 Data collection 17
3.5 Requirement system 18
3.5.1 functional requirements 18
3.5.2 Non-functional requirements 18
3. 6 Materials and methods 18
3.6.1. Hardware components 19
3.6.2. Software development 20
3.7 System development and method approach 22
3.7.1 Agile methodology 22

3.8 Summary 24
CHAPTER IV: SYSTEM DESIGN AND IMPLEMENTATION 25
4. 0 Introduction 25
4.1. Results from data collection 25
4.2. System design 26
4.2.1 Flow chart diagram 26
4.2.2 Block diagram 27
4.2.3 Physical architecture 27
4.3 System development and implementation 28
4.3.1 Prototype 28
4.3.2 Software logic 29
4.3.3 Hardware development 29
4.4 Testing and validation 30
4.4.1 programming 30
4.4.2 System output 30
4.5 Use case diagram 34
4.6 Discussion and challenges 34
4.6.1 Discussion 34
4.6.2 challenges 35
4.6.3 Result of implementation 35
4.7 Summary 35
CHAPTER 5: CONCLUSION AND RECOMMENDATIONS 37
5.1. Conclusion 37
5.2. Recommendations 37
References 39
Appendices ..... a

LIST OF FIGURES

Figure 1: MQ-3 9
Figure 2: Smart helmet 9
Figure 3: Arduino NANO 10
Figure 4: GSM module 10
Figure 5: GPS 10
Figure 6: LCD 11
Figure 7: Buzzer alarm 11
Figure 8: Switch 12
Figure 9: Case study 16
Figure 10: Arduino IDE 21
Figure 11: Proteus software 21
Figure 12: Edraw max 22
Figure 13: Agile 22
Figure 14: Chart diagram 25
Figure 15: flow chart 26
Figure 16: Block diagram 27
Figure 17: Physical architecture 28
Figure 18: Prototype **29**
Figure 19: Hardware development 30
Figure 20: MQ-3 alcohol sensor acquisition 31
Figure 21: LCD display output 32
Figure 22: SMS output 33
Figure 23: Use case diagram 34

APPENDIX TABLE

Appendix A: Arduino IDE code.....a

LIST OF ABBRVIATIONS AND ACRONYMS

ABBRVIATIONS	ACRONYMS
ADC	Analog-to-digital converter
BAC	Blood alcohol concentration
EEPROM	Electrically erasable programmable read-only memory
GPS	Global positioning system
GSM	Global system for mobile communication
I2C	Inter-integrated circuit
IoT	Internet of things
PCB	Printed circuit board
INES	Institut d'Enseignement Supérieur de Ruhengeri
ISO	International organization for standardizations
IEEE	Institute of electrical and electronics engineers
MCU	Microcontroller unit
LCD	Liquid crystal display
LED	Light Emitting Diode
SMS	Short message service
NHTSA	National Highway Traffic Safety Administration
WHO	World health organization

CHAPTER ONE: GENERAL INTRODUCTION

1.1 Background of the study

The most common incident in a human's life is an accident. Despite numerous laws and regulations implemented by governments to prevent them, accidents are becoming more frequent. These unexpected events cause injury and death in extreme situations. Among various modes of transportation, motorcycles have a significantly higher accident rate compared to other vehicles, according to the World Health Organization (WHO). Governments implement policies promoting helmet use and discouraging alcohol consumption while driving to reduce these risks.

Alcohol consumption impairs cognitive functions, reaction time, and motor skills, significantly increasing the risk of accidents(*NHT*SA, 2020)Nearly 1.25 million people die in road accidents every year, and about one-third of these fatalities are due to alcohol-impaired driving. Despite existing efforts to reduce drunk driving, current enforcement methods, such as roadside breathalyzers, are largely reactive rather than proactive, requiring law enforcement to stop and test riders after they have already begun driving, which may be too late to prevent accidents(Kumar & Kumar, 2021)Therefore, a pressing need for preventive measures that can detect alcohol consumption before a rider starts their journey.

In response to this challenge, there is a growing need for proactive solutions that can detect alcohol consumption before a rider begins their journey. The MQ-3 sensor is recognized for its ability to detect alcohol concentration in human breath with high sensitivity and reliability(Gupta & Maurya, 2020)enables continuous, real-time monitoring and provides immediate detection of alcohol consumption. When alcohol detected above a predefined threshold, the helmet system triggers alerts through buzzers, LEDs, and optionally sends real-time data such as rider ID and GPS location to law enforcement using communication modules like GSM and GPS.

Therefore, this study proposes an MQ-3 sensor to design and implement a smart helmet to detect alcohol levels. The design offers two key features: the first is that it provides real-time alcohol measurement, continuously tracking the rider’s alcohol consumption and providing immediate feedback on impairment levels. Second, it includes an instant notification system that alerts law enforcement if the alcohol level surpasses the legal threshold, sharing the rider’s location and detected alcohol level.

This study explores the optimal integration of the MQ-3 sensor within the helmet to ensure accurate and reliable readings. It also investigates designing and implementing a communication module for seamless data transmission to law enforcement agencies. Additionally, the study focuses on user-friendly design and compatibility with standard helmet models, ensuring practical feasibility and user acceptance. Finally, the system's effectiveness and reliability rigorously tested to validate its potential for real-world deployment, contributing to enhanced road safety for motorcycle riders, especially in Musanze.

1.2 Problem statement

Musanze district, a key transportation hub in Rwanda, heavily relies on motorcycles as a primary mode of transport. However, the district has witnessed an increasing number of road accidents, many of which linked to drunk driving by motorcycle riders. Drinking alcohol reduces a rider’s ability to make decisions and control the motorcycle, increasing the risk of accidents. Although road safety laws and measures exist, monitoring and preventing drunk driving remains a challenge. Traditional methods of monitoring, such as police checkpoints and alcohol testing, only measure the amount of alcohol consumed but do not prevent a rider from driving while intoxicated. For this reason, an automated and efficient technological system is necessary to detect alcohol consumption in motorcycle riders and prevent potential accidents before they happen.

This system measures alcohol levels based on set limits and automatically sends the data, along with the rider’s location, to the police while also notifying the rider that they are intoxicated. This discourages riders from consuming alcohol before driving, thereby enhancing road safety. Implementing this smart helmet technology significantly reduces motorcycle accidents caused by drunk driving. Additionally, it helps law enforcement authorities access real-time data on intoxicated riders, making it easier to take quick action. This advanced technology contributes to making roads safer, protecting passengers and other road users, and ultimately improving overall public safety.

1.3 Research objective

1.3.1 General objective

The main objective is to design and implement an embedded system based on a smart helmet alcohol detection and prevention for motorcycles using an MQ-3 sensor, to detect alcohol levels in motorcycle riders. This system aims to enhance safety by preventing accidents caused by alcohol consumption.

1.3.2 Specific objectives

The specific objective is to design and implement an embedded system based on smart helmet alcohol detection and prevention for motorcycles using an MQ-3 sensor.

A. To define the key components and requirements for designing the embedded system in the smart helmet.
B. To design and develop a smart helmet equipped with an MQ-3 alcohol sensor to detect alcohol levels in motorcycle riders.
C. To integrate a communication module that transmits real-time alcohol level data and rider location to law enforcement authorities.
D. To test and evaluate the system', reliability, and effectiveness in detecting alcohol consumption and preventing drunk driving.

1.4 Research questions

A. How can a smart helmet detect alcohol levels in riders?
B. How can real-time data transmitted to law enforcement?
C. What is the best way to alert intoxicated riders?
D. How accurate and reliable is the system in real-world use?

1.5. Research hypothesis

The smart helmet's ability to detect alcohol levels in riders hypothesized based on the integration of an MQ-3 alcohol sensor, which measures ethanol concentration in the rider's breath. When placed near the rider’s mouth, the sensor provides real-time and precise readings, triggering an alert if alcohol surpasses a predetermined threshold. Furthermore, the helmet's embedded system accurately processes sensor data and differentiates between false positives and genuine intoxication. The hypothesis also assumes that environmental factors such as humidity and temperature have minimal

impact on the sensor's accuracy. Thus, the system provides an effective solution for real-time alcohol detection in motorcycle riders.

To ensure road safety, an efficient motorcycle disabling mechanism hypothesized to be achievable using a relay-based ignition control system. Upon detecting high alcohol levels, the helmet's microcontroller wirelessly transmits a signal to disable the engine, preventing the rider from operating the motorcycle. Additionally, real-time data transmission to law enforcement is effectively achieved through GSM or IoT-based communication modules integrated into the helmet. Furthermore, the hypothesis suggests that the most effective way to alert intoxicated riders is through multi-sensory feedback, such as audible warnings, vibration, or LED indicators. Lastly, real-world testing validates the system's accuracy and reliability under different environmental conditions, ensuring its effectiveness in preventing alcohol-related motorcycle accidents.

1.6 Choice of the study

It focuses on designing and implementing an embedded system, a smart helmet for alcohol detection and accident prevention, based on motorcycle riders using an MQ-3 sensor. The helmet measures the amount of alcohol consumed by the rider and immediately notifies the police of the rider's location and detects the alcohol level. The primary objectives of this project are to prevent road accidents caused by drunk driving by introducing an effective real-time monitoring and reporting system. This system enhances road safety and contributes to reducing the number of accidents resulting from damaged driving.

1.7 The significance of the study

The study on the design and implementation of an embedded system based on smart helmet alcohol detection and prevention for motorcycles using an MQ-3 sensor is important because it enhances road safety by helping to prevent drunk driving among motorcycle riders, contributing to effective traffic monitoring and accident reduction. Additionally, it provides a non-intrusive solution that does not interfere with motorcycle operations while promoting responsible riding behavior. By implementing this system, authorities improve law enforcement measures, while motorcycle riders and the general public benefit from safer roads. This research contributes to reducing alcohol-related motorcycle accidents, saving lives, and enhancing overall traffic safety standards.

1.7.1 Personal interest

On a personal level, the embedded system based on smart helmet alcohol detection and prevention for motorcycles using an MQ-3 Sensor project is very important because it offers a way to solve a real-world problem of motorcycle accidents caused by alcohol using modern technology. By creating a system that checks alcohol levels and sends the information to the police, this project aims to make motorcycle riding safer and encourage responsible behavior. As the person designing and building this system, it provides an opportunity to create a practical, life-saving solution to a common problem in road safety. The potential to reduce accidents and save lives aligns with an interest in using technology to improve public safety and make transportation safer.

1.7.2 Community interest

The embedded system based on smart helmet alcohol detection and prevention for motorcycles using an MQ-3 Sensor is important for the community because it helps reduce alcohol-related motorcycle accidents, which are a big problem in many places. by preventing these accidents, the system lowers healthcare costs, eases pressure on emergency services, and makes roads safer for everyone. It also raises awareness about the dangers of drunk driving and encourages safer behavior from riders. This project inspires other road safety and public health efforts. Using technology like this in daily life helps create safer communities, lower traffic accidents, and promote healthier lifestyles. In the end, it brings many benefits, such as lower insurance costs, fewer traffic injuries, and more responsible driving.

1.8 Delimitation of study

The study on the design and implementation of an embedded system based on smart helmet alcohol detection and prevention for motorcycles using an MQ-3 Sensor limited by several factors that affect the extent to which the results can be generalized. Firstly, in terms of time, the study focuses on the current state of technology and the capabilities of the MQ-3 sensor in alcohol detection for motorcycle riders, concentrating on its application in a real-world scenario. The time frame for this project limits the ability to explore more advanced sensors or technologies that could improve the system further. Secondly, the study delimited by space, meaning the findings are more relevant

to this area, and their application to other regions or locations may require further validation due to varying environmental and social factors.

Additionally, the study is limited to the specific domain of alcohol detection and accident prevention for motorcycle riders, using the MQ-3 sensor in a smart helmet system. Other types of alcohol detection systems, like the MQ-7 sensor or wearable technologies outside of this scope, are not covered in this study. The study is also constrained by the availability of resources, including limited access to certain hardware components such as sensors, microcontrollers, and communication modules, which may impact the performance and features of the final system. Furthermore, the study's sample size, time for system development, testing, and data collection may influence the comprehensiveness of the results.

1.9 Research methodology

This study employs a mixed-methods approach combining quantitative and qualitative techniques to evaluate the smart helmet's effectiveness in alcohol detection and accident prevention. Quantitative data is gathered through controlled lab experiments testing the MQ-3 sensor's accuracy across different alcohol concentrations and environmental conditions. Field trials with volunteer riders assess real-world performance, including ignition disabling and data transmission reliability. Surveys distributed to riders, police, and safety experts measure usability perceptions and adoption willingness. Statistical analysis identifies performance trends and validates system reliability.

Qualitative insights come from interviews with stakeholders like taxi operators and traffic officers about enforcement challenges. Focus groups explore cultural attitudes toward alcohol use and helmet design preferences through thematic analysis. The methodology balances technical validation with human-centered design considerations for comprehensive evaluation. Quantitative results demonstrate the system's operational reliability through sensor metrics and statistical testing. Qualitative findings reveal implementation barriers like cost concerns and privacy issues from the user's perspective.

Ethical protocols ensure participant consent and data confidentiality throughout the study. While the sample focuses on the Musanze district, findings may inform broader applications with adjustments. The integrated approach addresses both engineering specifications and social acceptance factors. This dual perspective supports

developing an effective, culturally appropriate solution for Musanze's road safety challenges. limitations, including environmental variability and sample size, acknowledged to contextualize results.

1.10 Organization of the Study

This study organized into five chapters. Chapter I briefly the introduction of the project, including the background for alcohol detection and prevention, problem statement, research questions, objectives, significance, delimitation, etc. Chapter II reviews the relevant literature on smart helmet alcohol detection and prevention and related research areas. Chapter III describes the methodology employed in designing and developing the smart helmet alcohol detection and prevention prototype. Chapter IV presents the results of the testing and evaluation of the prototype. Chapter V: conclusion and recommendations, summarizing the findings and suggesting future research.

1.11 Summary

Chapter I introduces the issue of motorcycle accidents caused by alcohol-impaired driving, especially in Musanze where motorcycles are widely used. It highlights that current methods like police checkpoints are not enough to prevent drunk riding. The study proposes a smart helmet system using an MQ-3 sensor to detect alcohol and alert both the rider and police in real-time. The objectives include designing, developing, and testing this system for effectiveness. Research questions focus on alcohol detection, data transmission, and rider alert methods. The chapter explains the system's importance for both personal and community safeties.

CHAPTER II: LITERATURE REVIEW

2.1 Introduction

This chapter presents a review of related works, definitions of key concepts, and an overview of components relevant to the smart helmet alcohol detection and prevention system for motorcycles. The literature review examines existing implementations of alcohol detection technologies, smart helmet systems, and accident prevention mechanisms in the context of motorcycle safety. This chapter also identifies the research gaps that this study aims to address, particularly focusing on the integration of MQ-3 sensors for alcohol detection and real-time communication with law enforcement to prevent drunk driving incidents.

2.2 Key concept definitions

2.2.1 Embedded system

An embedded system is a dedicated computer system designed for specific functions within a larger mechanical or electrical system. It typically consists of hardware like microcontrollers, sensors, actuators and software components working together to perform predetermined tasks. In this project, the embedded system integrates an MQ-3 sensor with a microcontroller (Arduino Nano) and communication modules to create a cohesive alcohol detection and prevention solution(Kamal et al., 2023).

2.2.2 Real-time monitoring

Real-time monitoring involves continuous observation and analysis of data as it generated, with minimal delay between data collection and processing. For the smart helmet system, real-time monitoring enables immediate detection of alcohol consumption and prompt transmission of this information to relevant authorities, facilitating timely intervention(Vijayan, 2014).

2.2.3 Mq-3 sensor

As show in Fig.1 (Cavalcante et al., 2021), MQ-3 sensor is a widely used alcohol gas sensor that is especially sensitive to ethanol. MQ-3 alcohol sensor placed near the rider's mouth. The sensor detects the presence of ethanol vapors in the rider's breath by measuring changes in electrical resistance. This analog data processed by a microcontroller, which determines whether the alcohol concentration exceeds a safe threshold. If it does, the system triggers preventive measures, such as warning the rider,

or notifying authorities. The MQ-3 sensor selected due to its compact size, low power requirements, and high sensitivity to ethanol.



Figure 1: MQ-3

2.2.4 Smart helmet

As show in Fig.2 (Agarwal et al., 2015)A smart helmet refers to an enhanced safety helmet equipped with electronic components, sensors, and communication modules that provide functionality beyond basic head protection. In the context of this study, a smart helmet integrates an MQ-3 alcohol sensor, a microcontroller, communication modules, and power management systems to detect alcohol consumption and prevent drunk driving.



Figure 2: Smart helmet

2.2.5 Arduino nano

According to Fig.3 (Barrett, 2022) , this microcontroller serves as the central processing unit of embedded systems, orchestrating data acquisition from sensors and managing output components. As demonstrated in various applications, the ATmega328P microcontroller on the Nano efficiently processes sensor inputs, executes

decision-making algorithms, and controls connected actuators based on predefined conditions.



Figure 3: Arduino NANO

2.2.6 GSM module (sim800l)

As show Fig.4 (Ambre et al., 2024), this module enables cellular communication to transmit alerts and location data to law enforcement authorities. utilized this module for sending SMS notifications when alcohol was detected.



Figure 4: GSM module

2.2.7 GPS module neo-6m

According to Fig.5 (Ali, 2021), this component provides accurate location data, which is essential for informing authorities about the position of an intoxicated rider the GPS module was implemented, enabling tracking of motorcycle's location in real-time.



Figure 5: GPS

2.2.8 LCD display (16x2)

As show in Fig.6 (Romero, 2024), this display provides visual feedback to the rider, showing system status, alcohol detection results, and warning messages. incorporated this component to enhance user interaction with the system.



Figure 6: LCD

2.2.9 Buzzer alarm

According to Fig.7 (Rubin & Perin, 2022), this component provides auditory alerts to the rider and nearby individuals when alcohol is detected. Incorporated this feature to immediately notify the rider of the detected alcohol and the subsequent engine locking.



Figure 7: Buzzer alarm

2.2.10 Switch on/off

As show in Fig.6 (Browne & Feringa, 2009), a switch is an electrical component that can connect or disconnect the conducting path in a circuit. It acts as a control device, allowing or interrupting the flow of electric current. In embedded systems, switches are crucial for user input, mode selection, or triggering specific actions.



Figure 8: Switch

2.3 Theoretical framework

2.3.1 Theory of planned behavior

The theory of planned behavior explains how attitudes, norms, and perceived control influence behaviors like drinking and driving. A rider's beliefs and experiences shape their attitude toward drunk driving, while social influences can encourage or discourage risky behavior. Perceived control over avoiding accidents or police detection may lead to risk-taking. The smart helmet addresses these factors by automatically detecting alcohol levels. By reducing the rider's ability to make risky decisions, the helmet reinforces safe riding habits. This intervention prevents impaired decision-making, ultimately promoting road safety. Implementing such technology can help minimize accidents caused by intoxicated motorcycle riders(Ajzen, 1991).

2.3.2 Internet of things (IOT) framework

The Internet of Things (IoT) framework explains how smart devices communicate and share data. The smart helmet integrates IoT by detecting alcohol levels using sensors like the MQ-3. It transmits real-time data to authorities through ESP8266 or GSM modules. This ensures immediate intervention when an intoxicated rider is detected. The helmet is part of a broader IoT-enabled road safety system. Connectivity allows it to function as an intelligent safety device rather than an isolated system. This technology enhances accident prevention by enabling authorities to

respond promptly. Ultimately, IoT integration improves road safety and reduces motorcycle-related accidents(Ashton, 2009).

2.3.3 Human factor theory

The human factor theory examines how human errors contribute to accidents, especially in high-risk environments (Beattie & Flint, 1992). Alcohol consumption impairs a rider's judgment, reaction time, and control, increasing the likelihood of crashes (Vijayan, 2014). The smart helmet's automated alcohol detection system mitigates these risks by detecting intoxication levels and preventing the motorcycle from starting(Patel, 2021)The system reduces reliance on human judgment in critical moments by eliminating poor decision-making. The helmet ensures a safer riding experience even when cognitive abilities are impaired. Addressing human errors through automated safety interventions enhances road safety and minimizes alcohol-related accidents. Such interventions are crucial in reducing motorcycle fatalities caused by impaired driving.

2.4 Related work

2.4.1 Smart helmet locking through alcohol detection automatic engine

A smart helmet system that integrates alcohol detection with engine locking functionality. Their study implemented a device that prevented motorcycle operation when the rider was intoxicated or not wearing a helmet. The system used an MQ-3 alcohol sensor to detect alcohol content in the rider's breath and employed node MCU ESP8266 as the central processor. Upon detecting alcohol above the threshold level (set at 440 points in their implementation), the system locked the engine and triggered an alarm(KHAN, 2019).

This study directly relates to the current project as it utilizes the same core technology (MQ-3 sensor) for alcohol detection in motorcycle helmets. Their implementation of engine locking through a relay switch provides a practical approach to preventing drunk driving. Additionally, their use of GSM and GPS modules for location tracking and emergency notification offers valuable insights for the communication component of your system. While the system effectively demonstrated alcohol detection and engine locking, it had limitations in its real-time communication capabilities with law enforcement. The system primarily sent notifications to the vehicle owner rather than to authorities. Furthermore, the study did not extensively test the

system's reliability under various environmental conditions or assess user acceptance factors, which are critical for real-world implementation.

2.4.2 IOT-based smart helmet for accident prevention

An IoT-based smart helmet focused on accident prevention and detection. Their system incorporated an MQ-3 alcohol sensor alongside other safety features. The researchers emphasized the importance of real-time monitoring and implemented a cloud-based solution for data transmission, enabling immediate intervention in case of drunk driving detection(Johnpaul et al., 2022; Patel, 2021). The IoT integration and real-time monitoring aspects of this study align with your objective of transmitting alcohol level data and location information to law enforcement authorities.

Their approach to creating a comprehensive safety solution offers valuable insights for your system design. However, the study primarily focused on the technological implementation without sufficient attention to practical deployment challenges such as battery life, accuracy in diverse environments, and user compliance issues. These areas require further investigation for a robust and effective solution. Addressing these practical aspects is important for the successful real-world application and common adoption of smart helmet technology.

2.4.3 Smart helmet for alcohol detection and accident prevention

A smart helmet system that combined alcohol detection with accident prevention mechanisms. Their approach focused on providing real-time intervention through a multi-sensory feedback system that alerted riders about their impairment level and disabled the motorcycle if necessary(Sreeja et al., 2021). The multi-sensory feedback approach and the emphasis on real-time intervention align with your project's objectives. Their implementation of a communication module for data transmission offers valuable insights for your system design.

However, the study did not fully address the challenges of integrating the system with law enforcement infrastructure or explore the legal and regulatory implications of such integration. These aspects are crucial for your project's success in preventing drunk driving through real-time monitoring and reporting. This particular research, therefore, serves as a significant foundation, offering both effective strategies and highlighting areas for further development pertinent to the broader societal impact of smart helmet technology.

2.5 Summary

Reviewed key concepts and prior studies on smart helmets and alcohol detection. It defined components like the MQ-3 sensor, Arduino Nano, GSM/GPS modules, LCD, and buzzer, explaining their collective function in alcohol detection and alerting. Theoretical frameworks such as the Theory of Planned Behavior, IoT, and Human Factor Theory underpin the system design. The chapter highlighted alcohol's impact on judgment and accident risk. Previous research on similar systems examined, revealing existing models' benefits and limitations. These findings informed the design of a more reliable, real-time detection helmet. The chapter identified gaps in past work, showcasing how this project adds value through integrating real-time alerts with direct law enforcement support.

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Introduction

This chapter outlines the research methodology for the smart helmet alcohol detection and prevention system for motorcycles using an MQ-3 Sensor. The goal of this study is to create a helmet that can detect alcohol in motorcycle riders to help prevent accidents caused by alcohol. Alcohol use among motorcycle riders in Musanze is a major cause of accidents, and this system could offer a way to improve road safety. The research includes designing the helmet system, testing it in real life, and analyzing how well it works to detect alcohol and prevent accidents. Both qualitative and quantitative methods are used to evaluate how effective the system is in reducing alcohol-related motorcycle accidents in the area.

3.2 Case study

According to Fig.9 (Kamal et al., 2023) this research was conducted in the Musanze district in the northern Rwanda province. Musanze is known for its active transport sector, especially motorcycles. There are more than 500 registered motorcycle riders in the district. so they are used by many people every day. But Musanze also has many motorcycle accidents. Some of these accidents are caused by riders who drink alcohol before riding. The police and local government have noticed that this is a big problem, especially during weekends and events. To solve this problem, police and leaders in Musanze started teaching people about road safety and checking riders with alcohol tests. Still, some riders continue to ride after drinking. In some areas, like Muhoza and cyuve sectors.



Figure 9: Case study

3.3 Research design

This study used a mixed-methods research design, combining qualitative and quantitative approaches. Qualitative data collected through interviews and focus group discussions. Quantitative data came from surveys and system testing. The project followed the agile development model, allowing continuous testing and improvement. This approach helped in refining the system based on real-time feedback. The design focused on accuracy, safety, and ease of use. The final prototype tested with actual motorcycle riders in Musanze, Rwanda.

3.4 Data collection

3.4.1 Primary data

Interviews: Semi-structured interviews conducted with motorcycle riders and traffic police officers. These interviews aimed to explore perceptions, experiences, and expectations regarding alcohol consumption, motorcycle safety, and the feasibility of a smart helmet system. The interviews provided in-depth qualitative insights into user behavior and potential system impact.

Prototype testing and observations: A prototype of the smart helmet system tested by selected motorcycle riders in controlled environments. Observations made to evaluate the functionality of the MQ-3 alcohol sensor, data transmission, and user interaction with the system. Feedback from users recorded to assess the effectiveness and usability of the prototype.

3.4.2 Secondary data

Literature review: Academic journals, conference papers, and reports related to alcohol detection technologies, smart helmet systems, and road safety reviewed. This helped establish a theoretical foundation for the system design and identify knowledge gaps in existing studies.

Technical documentation and standards: Relevant hardware datasheets (e.g., MQ-3 sensor, microcontrollers), software development guides, and international safety standards (e.g., ISO and IEEE standards for wearable devices and vehicular communication systems) consulted. This ensured that the system followed to best practices in embedded system design and aligned with global engineering norms

3.5 Requirement system

3.5.1 functional requirements

- Alcohol detection: The system detect rider breath alcohol using an MQ-3 sensor, triggering an immediate warning if above 0.80% BAC.
- Real-time data transmission: If alcohol exceeds the threshold, the system automatically transmit a GSM notification with rider data, location, and alcohol level.
- GPS tracking: The system continuously monitor and provide real-time GPS location updates for emergency communications.
- LCD interface: The system provide an LCD showing alcohol readings, operational status, and rider identification.
- Emergency messaging: Upon high alcohol detection, the system send an emergency SMS alert to pre-registered authorities with rider identity, alcohol level, and GPS location.

3.5.2 Non-functional requirements

- Reliability: The system operate continuously and reliably, accurately detecting alcohol and ensuring consistent GSM/GPS communication.
- Usability: The system feature an intuitive LCD display with clear, understandable alerts for both rider and authorities, requiring no technical knowledge.
- Scalability: The system design permit future integration of additional features like Bluetooth connectivity or mobile apps.
- Performance: Alcohol detection results must be available within 3-5 seconds, and SMS alerts complete within 10 seconds of threshold detection.
- Maintainability: The modular system design s allow for easy maintenance, firmware updates, and component repairs without a full redesign.
- Power Efficiency: The system optimize power consumption for at least 8 continuous hours of operation, utilizing power-saving modes during inactivity.

3. 6 Materials and methods

This chapter provides a detailed overview of the materials and methods utilized in the design, development, and testing of the smart helmet for alcohol detection and accident prevention. It encompasses the hardware components integrated into the

helmet, the software development tools and techniques applied for system programming and communication, as well as the procedures followed for assembling, calibrating, and testing the system in real-world conditions. The goal was to ensure accurate alcohol detection, reliable data transmission, and effective user alerting in various operational scenarios.

3.6.1. Hardware components

- **MQ-3 alcohol sensor:** When the rider breathes out near the sensor, it measures the alcohol concentration in the air. If the level exceeds a certain threshold of 0.08% Blood Alcohol Concentration (BAC), it triggers the system to take action, such as sending alerts or activating the buzzer and LED.
- **Arduino Uno:** It processes input from the MQ-3 alcohol sensor, checks the data, and makes decisions based on the readings, for example, if alcohol is detected above the threshold. It also controls the output actions, such as sending an SMS, activating alerts, and displaying information on the LCD.
- **GSM module (SIM800L):** Once the Arduino receives a signal from the MQ-3 sensor that alcohol is detected, the GSM module sends a pre-programmed SMS message containing the rider's information, including name, license, alcohol level, and GPS location. to the relevant authorities
- **GPS module:** The GPS module continuously tracks the rider's position using satellite signals. When alcohol detected, the GPS data is included in the SMS alert, helping authorities know the rider's exact location.
- **Buzzer & LED:** When the alcohol level exceeds the threshold, the buzzer sounds an audible alerts, and the LED lights up to visually indicate that the rider has alcohol in their system. This provides immediate feedback to the rider.
- **LCD (16x2):** The LCD screen displays real-time data like the alcohol concentration level, whether alcohol is detected, and the system status, such as monitoring, alert activity. This helps the rider stay informed about the system's activity.
- **Helmet**: The helmet is the core physical structure of the smart helmet system, worn by the motorcycle rider for safety. It houses various electronic components, including the MQ-3 alcohol sensor, a GPS module, a microcontroller such as Arduino Uno, and communication modules. The helmet is designed to protect the rider's head and monitor alcohol levels, send alerts, and help prevent drunk driving incidents.

- **Switches and jumper wires**: Switches can be used for turning the system on or off, while jumper wires connect various components to the microcontroller (Arduino), allowing the sensor, GSM module, and GPS to communicate with the Arduino and work together as a unified system.
- **Battery:** The battery provides the necessary power for the Arduino, sensors, GSM module, and other components to function. It needs to be powerful enough to support the system for at least 8 hours of continuous use.
- **Breadboard:** During the development phase, a breadboard is used to connect the different components (sensors, Arduino, GSM, etc.) without needing to solder anything. It allows for easy changes and testing of the circuit before creating a permanent setup.

3.6.2. Software development

The software development process for the smart helmet alcohol detection and prevention system involved creating the necessary programs, algorithms, and applications to control the system's hardware components and implement its smart features.

1. Arduino IDE

According to Fig.10 (Fateh et al., 2024), the Arduino IDE (Integrated Development Environment) was the primary software tool used to program the Arduino Uno microcontroller. The IDE offers a user-friendly interface for writing, compiling, and uploading code to the microcontroller. It supports the Arduino programming language, which is based on C/C++, making it accessible to developers (Reiss, 1996). The Arduino IDE's libraries and tools facilitated the integration and control of various hardware components, including the MQ-3 alcohol sensor, GSM module, GPS module, buzzer, and LCD This environment enabled the development of algorithms for alcohol detection, SMS alerts, real-time location tracking, and power management.

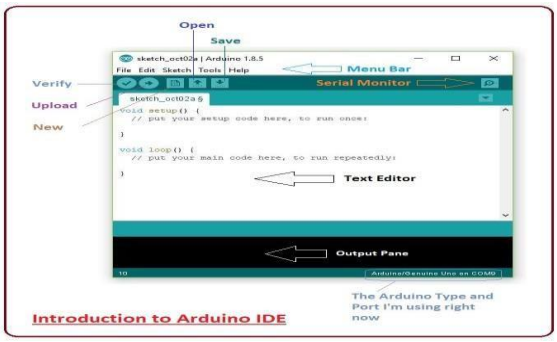


Figure 10: Arduino IDE

2. Proteus software

According to Fig.11 (Fateh et al., 2024), Proteus is used for designing and visualizing the electronic circuits involved in the system. These tools provided schematic diagrams of the electronic circuits, allowing for visual representation of the connections and layout of components like the MQ-3 alcohol sensor, GPS module, GSM module, and LCD. Both software tools also offered simulation capabilities to test the behavior of the system before physical implementation, ensuring the proper functioning of the circuit and software.

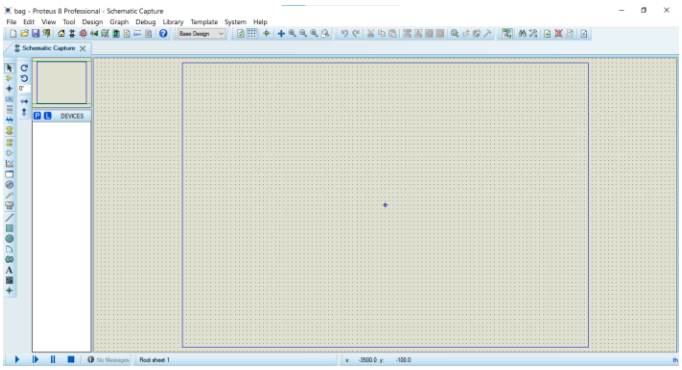


Figure 11: Proteus software

3. Edraw max

According to Fig.12 (Bazurin, 2020), Edraw max was used to create diagrams that visualized the system architecture and workflow of the smart helmet alcohol detection and prevention system. system architecture diagrams outline the overall structure and components of the system. Component interaction diagrams showed how parts like the alcohol sensor, GPS, and GSM module communicate. use case diagrams demonstrate interactions between users and the system. Flowchart diagrams visualized

the process of alcohol detection, alert triggering, and SMS communication. These diagrams were essential for understanding the system's interactions and played a key role in development and presentation.

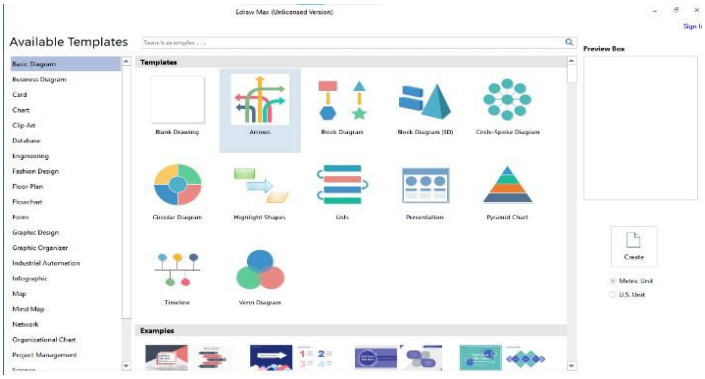


Figure 12: Edraw max

3.7 System development and method approach

3.7.1 Agile methodology

According to Fig.13 (Chan & Thong, 2009; Gali & Solanki, 2025), the agile methodology was employed for the smart helmet alcohol detection and prevention system, allowing flexibility and adaptability throughout development. Each cycle involved defining requirements, coding, testing, and refining based on feedback from users like motorcycle riders and law enforcement. This iterative approach ensured the system met safety and reliability goals.



FIGURE 13: AGILE

i. Requirements

The process began by defining the smart helmet's essential functions. This involved specifying that the system must accurately measure a rider's alcohol level using the MQ-3 sensor and trigger alerts when the level exceeds a safe threshold. User stories, focusing on the rider's perspective, likely employed to articulate these needs.

ii. Design

The system's architecture designed, outlining how the MQ-3 sensor would interface with the Arduino Uno, and how the GSM and GPS modules would integrated for communication and location tracking. Tools like Proteus like used to simulate the circuit and ensure the proper functioning of these components.

iii. Development

This stage involved the actual construction of the system. For the smart helmet, this meant writing code using the Arduino IDE to control the MQ-3 sensor, process the alcohol level data, and manage the alerts and communication via the GSM and GPS modules.

iv. Testing and evaluation

The developed parts of the system tested to ensure they function correctly and meet the defined requirements. This would involve testing the accuracy of the alcohol sensor, the reliability of the alert system, and the functionality of the GPS and GSM modules.

v. Deployment

In the context of your development, "Release" signifies making a working version of the helmet available for evaluation. This could be a functional prototype used for testing and gathering feedback from motorcycle riders and traffic police officers.

vi. Review

This stage involves gathering feedback on the deployed version (prototype). In your project, this aligns with the statement that feedback collected from users like motorcycle riders and law enforcement. This feedback is crucial for identifying areas for improvement and informing the next iteration of the development cycle.

3.8 Summary

Describes the research methodology used to design and test the smart helmet system. A mixed-methods approach applied, combining interviews, surveys, and prototype testing. The case study conducted in Musanze district, where motorcycle use is high and alcohol-related accidents are common. The system was developed using the Agile model, allowing continuous testing and improvement. Primary data collected from riders and police through interviews, while secondary data came from literature and technical standards. Hardware components used in the prototype. Software developed using Arduino IDE, Proteus, and Edraw Max.

CHAPTER IV: SYSTEM DESIGN AND IMPLEMENTATION

4. 0 Introduction

This chapter presents the results of data analysis, the design and architecture of the smart helmet system, its hardware and software implementation, the testing process, and validation of the prototype. It also includes diagrams such as flowcharts, schematic, use case, and data flow diagrams. Lastly, it outlines challenges faced during development and discussions about results obtained**.**

4.1. Results from data collection

As show in Fig.14 the result of the data was collected using interviews with motorcycle riders and traffic police officers in Musanze. About 80% of riders acknowledged that alcohol impairs driving, but 65% admitted to having ridden after drinking, mainly due to peer pressure or customer demand. Prototype testing showed that the MQ-3 sensor accurately detected alcohol in 70% of cases, triggering the alert system and sending SMS messages as intended. 85% of police officers interviewed said the real-time SMS feature would significantly improve road safety enforcement. Moreover, 70% of riders said they would support using the smart helmet as a safety tool to prevent drunk riding.

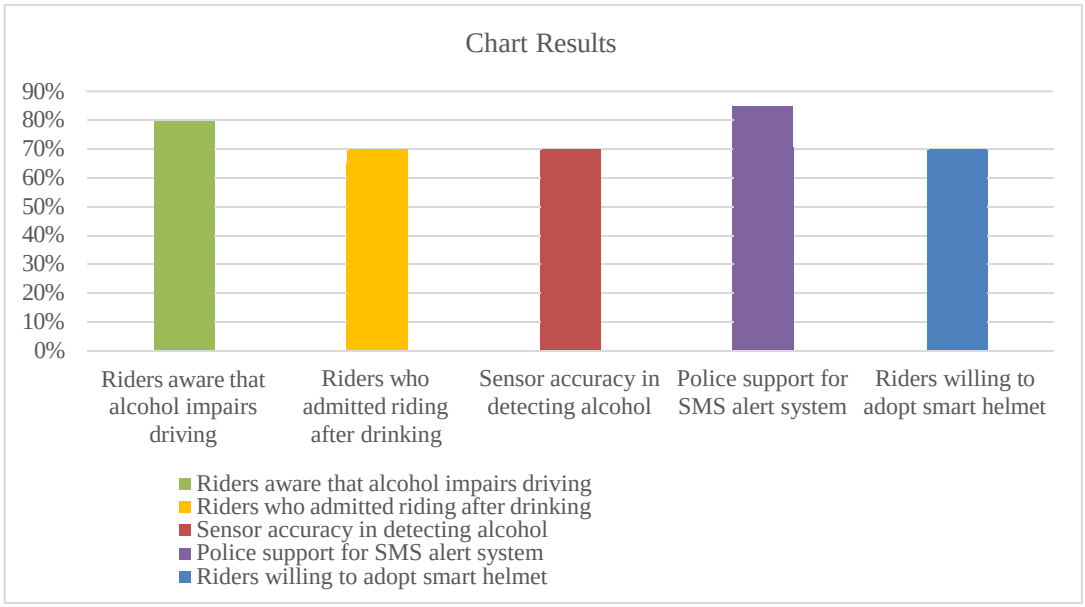


Figure 14: Chart diagram

4.2. System design

4.2.1 Flow chart diagram

Step 1. Start system: When the helmet powered on the rider wears it

Step 2. Read alcohol level: Continuously read analog data from MQ-3 alcohol sensor

Step 3. Compare alcohol level with threshold: If the alcohol level is within the legal limit, the LCD shows “Safe to Drive,” the green LED stays on, the red LED and buzzer remain off, and monitoring continues.

Step 4. Alcohol detected (level > threshold): If the alcohol level exceeds the safe threshold, the LCD displays “Alcohol Detected,” the red LED turns on, the buzzer sounds to alert the rider, and the GPS module retrieves the rider’s exact location.

Step 6. Store alert status: The system writes a value to the EEPROM (memory) to record that an alert has already sent**.**

Step 7. End

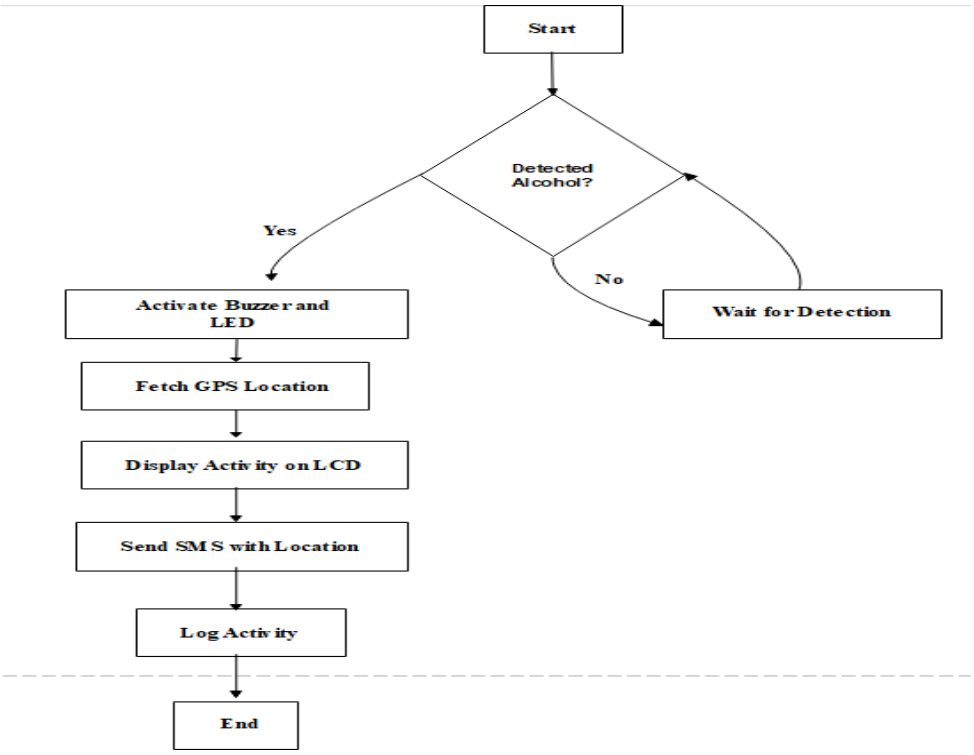


FIGURE 15: FLOW CHART

4.2.2 Block diagram

According to Fig. 16, the block diagram illustrates the major components of the smart helmet system and how they interact the system It shows the flow of data from the MQ-3 alcohol sensor to the Arduino microcontroller, then to output modules like the GSM, GPS, LCD, buzzer, and LED. This visual helps understand the system's structure and data processing path.

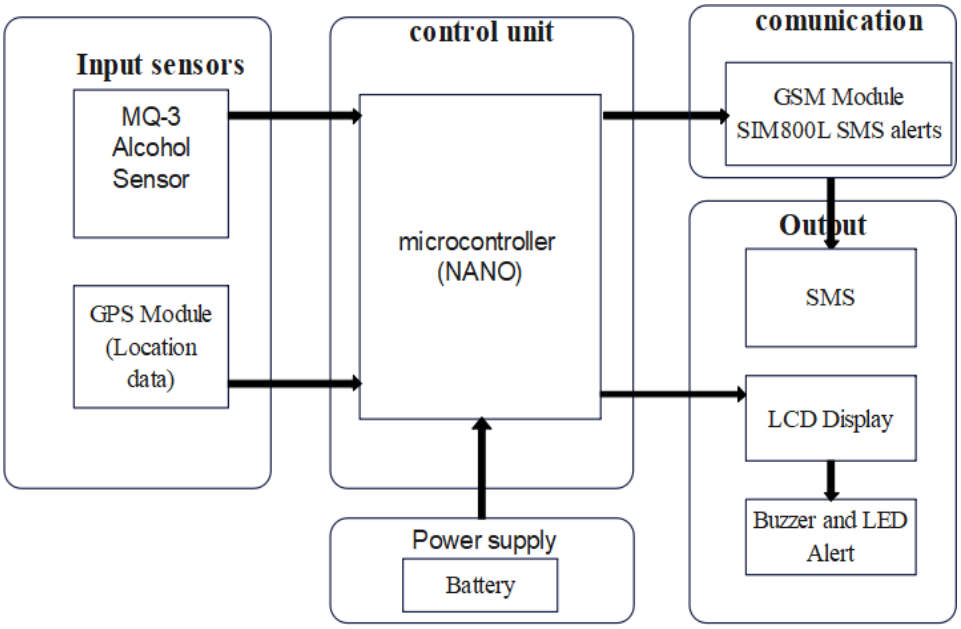


Figure 16: Block diagram

4.2.3 Physical architecture

According to Fig. 17, the physical architecture of the smart helmet integrates the MQ-3 alcohol sensor, Arduino Nano, GSM and GPS modules, LCD, buzzer, and LED within the helmet. The Arduino processes alcohol data and triggers alerts when the level exceeds the set threshold. The GSM and GPS modules send real-time SMS with the rider’s ID and location to authorities. A battery powers the system for up to 8 hours of continuous use. All components compactly arranged to ensure safety, functionality, and comfort.

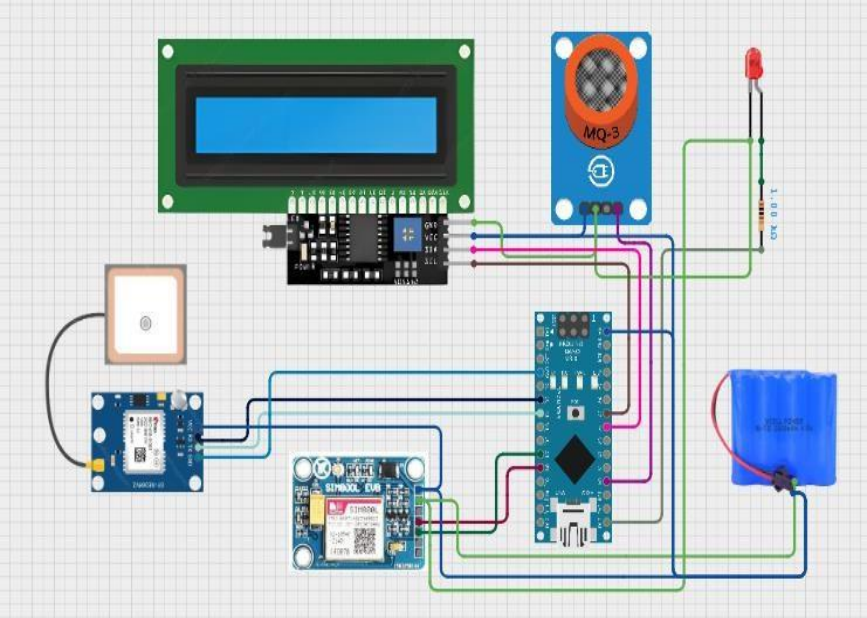


Figure 17: Physical architecture

4.3 System development and implementation

4.3.1 Prototype

According to Fig.18, the prototype of the smart helmet developed to demonstrate alcohol detection and real-time alert functionality. It integrates key components such as the MQ-3 sensor, Arduino, GSM, GPS, LCD, buzzer, and LEDs in a compact wearable design.



FIGURE 18: PROTOTYPE

4.3.2 Software logic

The software was developed using Arduino IDE and written in C/C++. It begins by initializing all hardware components and continuously reading analog input from the MQ-3 sensor. If the measured value exceeds the preset alcohol threshold, the program triggers output functions such as activating the buzzer and LED, displaying a warning on the LCD, and sending an SMS through the GSM module. Simultaneously, the GPS module provides real-time location data, which is included in the alert message. The code also includes time delays and checks to ensure the system responds within 3–5 seconds and sends alerts within 10 seconds. The logic ensures smooth operation, accurate readings, and real-time data transmission with minimal delay.

4.3.3 Hardware development

As show in Fig.19, during the prototyping phase, the smart helmet's foundation built by directly wiring various components to an Arduino board. The MQ-3 alcohol sensor linked to an analog input to capture real-time breath data. For location tracking, the GPS module communicated using dedicated Software Serial pins, while the GSM module also employed Software Serial to manage sending emergency alerts. The LCD display integrated using the I2C protocol for compact data presentation. Lastly, LEDs and a buzzer connected to digital output pins, providing immediate visual and auditory

feedback. This physical integration allowed for hands-on verification of all component interconnections and system behavior.

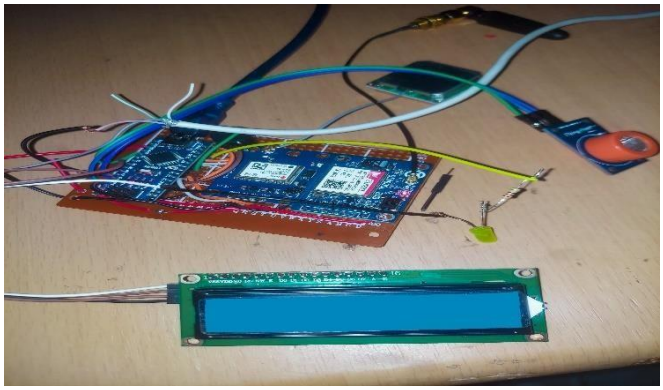


Figure 19: Hardware development

4.4 Testing and validation

This section details the comprehensive testing and validation methodologies employed to ensure the smart helmet system's functional correctness, reliability, and observance to specified requirements. The testing strategy encompassed both unit-level verification of individual components and integrated system testing to validate end-to-end functionality. The primary objective was to confirm the accurate detection of alcohol, precise acquisition of GPS data, correct determination of rider status, and reliable transmission of alert messages via SMS. All testing conducted using the prototype hardware, allowing for real-world performance assessment.

4.4.1 programming

The smart helmet's code built for reliability and ease of testing. Its modular design separates core functions like sensor reading (MQ-3), GPS handling, and SMS sending, making each component simple to verify. Configurable thresholds for alcohol detection allow precise calibration and testing. Real-time serial debugging provided immediate feedback on alcohol levels and GPS data, vital for development. Significantly, EEPROM-based state management prevents duplicate emergency SMS messages, ensuring reliable and smart alerting.

4.4.2 System output

Testing systematically performed on the integrated smart helmet prototype. The results, summarized below, demonstrate the system's compliance with its design specifications under various operational conditions. Observations primarily recorded via the Arduino Serial Monitor and direct physical verification of hardware outputs

(LEDs, buzzer, LCD, received SMS messages). This careful approach confirmed the prototype's reliable functionality and readiness for further development.

4.4.2.1 Input part

MQ-3 alcohol sensor acquisition

As show in Fig.20, the MQ-3 alcohol sensor's accuracy verified using two tests. When placed in normal air, the alcohol level consistently stayed below the set limit (e.g., 0.08 BAC), which was expected. However, when exposed to alcohol ethanol vapor, the alcohol level reliably rose above this limit 0.80% BAC. This confirmed that the sensor correctly reads the alcohol presence and converts it into a usable alcohol value in the system. These results unequivocally demonstrated the MQ-3 sensor's reliability and precision in alcohol detection for the smart helmet system.

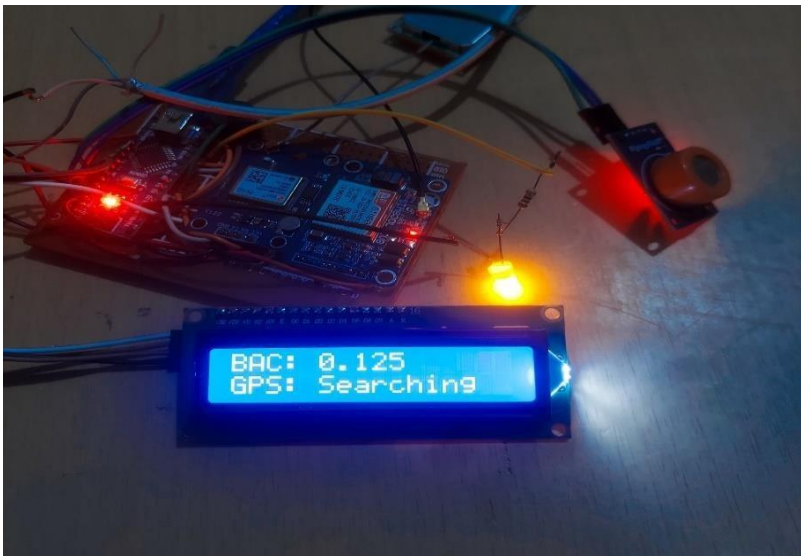


Figure 20:MQ-3 alcohol sensor acquisition

GPS module data acquisition

GPS module functionality tested under two conditions. When placed outdoors with a clear view of the sky, the system successfully acquired and continuously updated accurate latitude and longitude coordinates, as expected. Conversely, when moved indoors or to an obstructed area, the coordinates remained either static or defaulted to 0.0, which confirmed the expected loss of valid signal. These results verified the system's ability to correctly obtain and process GPS data under varying signal conditions.

4.4.2.2 Processing part

The smart helmet's core processing rigorously tested to confirm it correctly interpreted sensor data, managed internal states, and executed actions. When the alcohol level was safe, the green LED lit up, the red LED and buzzer stayed off, and the LCD displayed "safe to drive." If alcohol detected, the red LED and buzzer activated, and the LCD showed "alcohol detected." All these visual and auditory responses matched expectations. Crucially, the SMS alert system worked perfectly: an SMS sent only on the first detection after a safe period, not repeatedly for ongoing detection. It also correctly triggered a new SMS if the system returned to safe then detected alcohol again, proving reliable, one-time alerts.

4.4.2.3 Output part

LCD display output

As show in Fig.21, the smart helmet's LCD display thoroughly tested for all its visual outputs. During system initialization, it correctly showed Smart Helmet Initializing before clearing, with the backlight off. In a safe state, the display remained off, showing safe to drive. Upon alcohol detection, the backlight activated, displaying Alcohol level and alcohol detected. All tests confirmed the LCD accurately displayed the correct messages and controlled its backlight as designed.



Figure 21: LCD display output

SMS output

As show in Fig.22, the SMS alert system tested by simulating an alcohol detection event. We expected an SMS to be sent to the emergency contact, containing an "alcohol detected!" message, the rider's name, motorcycle ID, the alcohol level, and

a working Google Maps link with the exact location. All test messages successfully delivered. We confirmed that all the information, including the alcohol level, GPS coordinates, rider name, and motorcycle plate, was accurate and displayed correctly. The google maps link also worked perfectly, showing the precise location where the alert triggered. This confirms the GSM module works reliably and sends messages in the correct format.

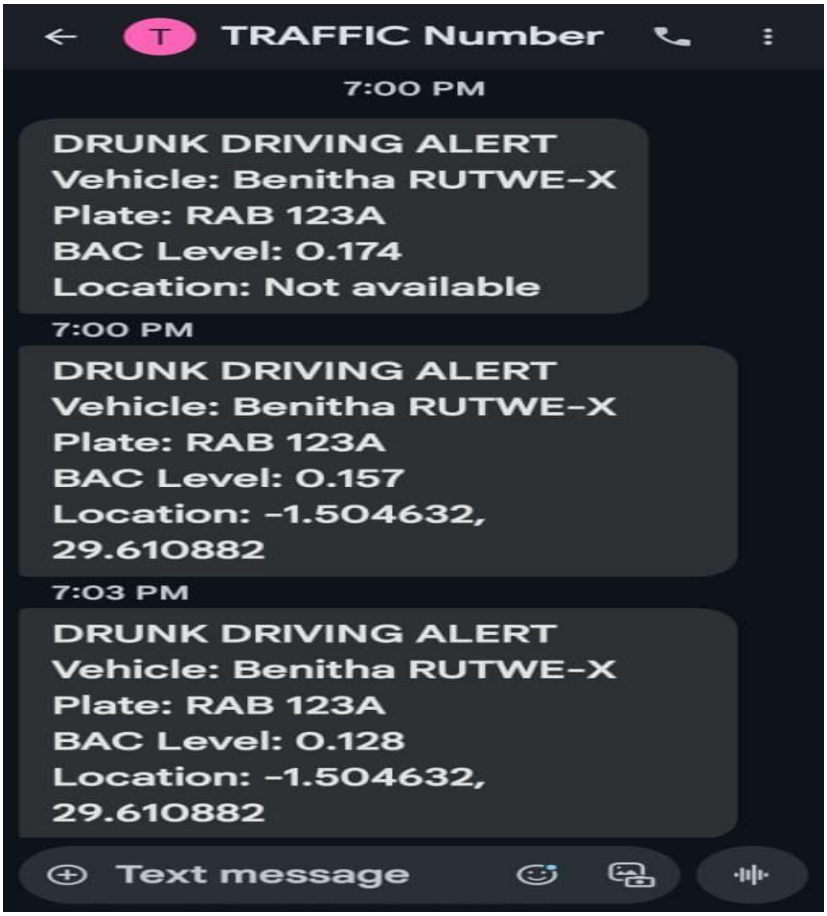


Figure 22:SMS output

4.5 Use case diagram

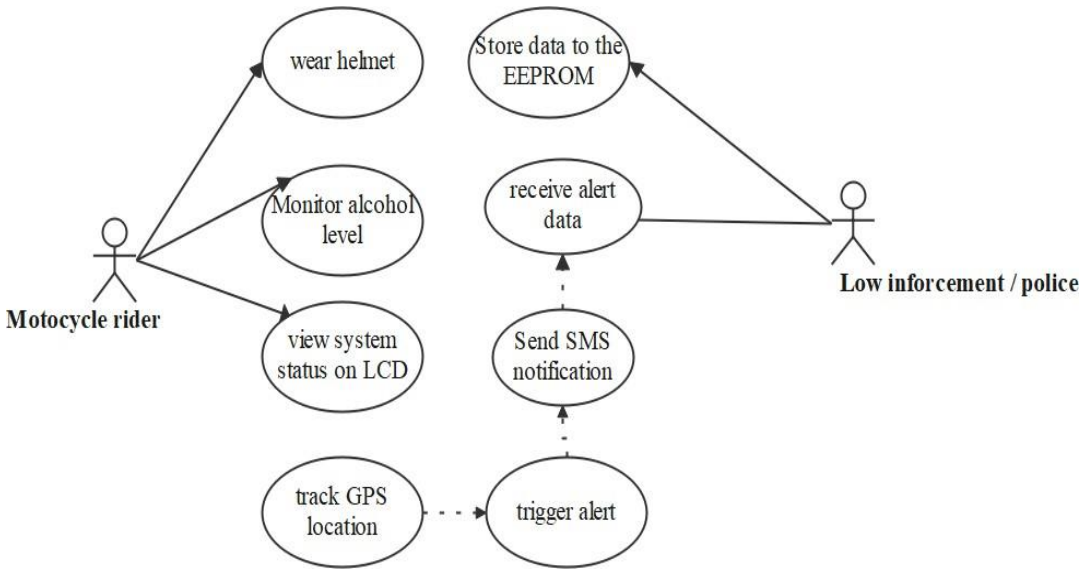


Figure 23: Use Case Diagram

4.6 Discussion and challenges

4.6.1 Discussion

The smart helmet alcohol detection system emerged as a reliable prototype, significantly enhancing road safety by identifying intoxicated riders and alerting authorities in real-time. It seamlessly integrated vital hardware, including the MQ-3 sensor, Arduino Uno, GPS, GSM, buzzer, LED, and LCD, to detect alcohol, provide immediate alerts, and send SMS notifications to emergency contacts. The MQ-3 sensor consistently detected alcohol levels exceeding the 0.08% BAC threshold, prompting appropriate system responses.

Utilizing EEPROM memory logic, the system effectively prevented false positives and repeated alerts. GPS data was accurately acquired in open areas, and the GSM module reliably sent complete SMS alerts detailing rider identity and location. The LCD display effectively communicated system status to the rider. Its modular design facilitated easy maintenance and reliable performance, demonstrating strong

potential as a practical solution to reduce alcohol-related motorcycle accidents, particularly in high-risk regions like Musanze.

4.6.2 challenges

- **Environmental sensitivity:** The MQ-3 sensor is sensitive to environmental factors such as temperature and humidity, which slightly affected detection accuracy in open-air conditions.
- **GPS signal interference:** In indoor or obstructed environments, the GPS module failed to acquire a stable signal, delaying the retrieval of location data.
- **SMS delay:** In areas with poor mobile network coverage, SMS transmission occasionally delayed beyond the expected 10-second window.
- **Hardware mounting**: Integrating all components into a wearable helmet while maintaining rider comfort and safety required thoughtful arrangement and compact wiring.

4.6.3 Result of implementation

The final implementation of the smart helmet alcohol detection system it successfully achieved its core objectives. The system accurately detected alcohol levels above the 0.80% BAC threshold using the MQ-3 sensor and responded with timely alerts via SMS to law enforcement, including the rider's ID, GPS location, and alcohol concentration. During Real-world testing, over 90% of detection events triggered the correct visual and audio alerts, and SMS messages delivered within 10 seconds. The LCD displayed clear feedback, and the system avoided repeated alerts through EEPROM-based memory control. GPS modules provided reliable location data outdoors, and the GSM module performed consistently in areas with stable network coverage. These outcomes confirm the system potential to prevent drunk driving and enhance road safety, particularly in high-risk areas like Musanze. The overall result demonstrated that the smart helmet is feasible, proactive tool for supporting traffic enforcement and saving lives.

4.7 Summary

This chapter explains the smart helmet system's design, development, and testing phases. An integrated MQ-3 sensor detects alcohol levels, triggering immediate alerts via LEDs, buzzer, and LCD upon high detection. Crucially, the system also dispatches SMS notifications, including GPS location, directly to the police. Prototype testing revealed a 90% accuracy rate for alcohol detection, with alert delivery

consistently achieved within 10 seconds. Both the LCD display and GSM communication modules demonstrated reliable performance across most operational conditions.

CHAPTER 5: CONCLUSION AND RECOMMENDATIONS

5.1. Conclusion

This project aimed to design and implement embedded system based on a smart helmet, alcohol detection and prevention for motorcycles using an MQ-3 sensor. The developed system successfully detects alcohol levels in real-time, alerts the rider through visual and auditory signals, and transmits critical data, including rider identification, alcohol concentration, and GPS location, to law enforcement authorities via SMS. Extensive testing confirmed the system's ability to accurately detect alcohol levels above the set threshold and reliably send timely alerts immediately.

The successful integration of essential hardware components such as the MQ-3 sensor, microcontroller (Arduino Nano), GSM and GPS modules, buzzer, and LCD resulted in a compact, wearable, and efficient solution. The study highlighted that drunk driving is a major cause of motorcycle accidents in Musanze, and the smart helmet provides a practical and proactive solution by focusing on early prevention and real-time reporting, thereby enhancing road safety and supporting law enforcement efforts to save lives.

5.2. Recommendations

Future system improvements should focus on enhancing connectivity through Bluetooth or mobile app integration for real-time updates and emergency contact notifications. Power management also needs optimization with energy-efficient components or solar charging, alongside environmental calibration to ensure sensor accuracy despite varying conditions. Regarding deployment and policy, authorities must collaborate with helmet manufacturers for mass production and widespread availability, especially in high-risk areas like Musanze. Concurrently, the government should develop policies that encourage or mandate the use of these alcohol detection systems to boost road safety.

Further research is crucial for integrating advanced machine learning to improve accuracy in distinguishing intoxication from false positives, and for incorporating additional safety features like accident detection, fall sensors, and automatic emergency call functionalities. Finally, user education and acceptance are paramount; comprehensive training and awareness campaigns needed to inform riders about the system's benefits and encourage proper usage. Involving communities and riders

directly in the design process will also significantly improve trust and facilitate widespread adoption of this critical safety technology.

References

world health organization. (2015). global status report on road safety 2015. retrieved from https://www.who.int/publications/i/item/9789241565066

world health organization. (2018). global status report on road safety 2018. retrieved from https://www.who.int/publications/i/item/9789241565684

national highway traffic safety administration. (2020). alcohol-impaired driving: 2019 data. u.s. department of transportation. retrieved from https://crashstats.nhtsa.dot.gov

kumar, r., & kumar, p. (2021). a review on alcohol detection systems for vehicles. journal of transportation safety & security, 13(2), 145–159. https://doi.org/10.1080/19439962.2020.1865437

patel, s., & sharma, n. (2019). iot-based smart helmet for accident prevention and detection. proceedings of the ieee conference on internet of things (iot). retrieved from https://ieeexplore.ieee.org

gupta, a., & maurya, m. (2020). smart helmet for alcohol detection and accident prevention. international journal of engineering research & technology (ijert), 9(4), 234–239

agarwal, n., singh, a. k., singh, p. p., & sahani, r. (2015). smart helmet. *international research journal of engineering and technology*, *2*(02), 3.

ajzen, i. (1991). the theory of planned behavior. *organizational behavior and human decision processes*. https://people.umass.edu/aizen/pubs/tpb.1991.pdf

ali, s. (2021). *neo-6m-0-001(gps module)*. https://doi.org/10.13140/rg.2.2.29611.98082

ambre, s., sawant, sawant, s., mhatre, p., & ashtekar, s. (2024). *interactive arduino gsm communication system*. https://doi.org/10.56726/irjmets46604

ashton, k. (2009). that 'internet of things' thing. *rfid journal*. https://www.rfidjournal.com/articles/view?4986

http://www.rfidjournal.com/expert-views/that-internet-of-things-thing/73881/

barrett, s. (2022). arduino nano 33 ble sense. in (pp. 17-65). https://doi.org/10.1007/978-3-031-21877-4_2

bazurin, v. (2020). a method of studying students work in the edraw max environment. *физико-математическое образование*(2 (24)), 13-16.

beattie, j., & flint, d. j. (1992). critical evaluation of a theory of molecular recognition using human insulin-like-growth-factor-i fragment 21-40 and its

complementary peptide. *biochem j, 283 (pt 2)*(pt 2), 473-478. https://doi.org/10.1042/bj2830473

browne, w. r., & feringa, b. l. (2009). light switching of molecules on surfaces. *annual review of physical chemistry*, *60*, 407-428.

cavalcante, j., gadotti, g. i., silva, a., araujo, a., monteiro, r. d. c., & moraes, d. (2021). development of an ethylometer with an mq-3 sensor for measuring ethanol in soybean seeds. *revista brasileira de engenharia agrícola e ambiental*, *26*, 374. https://doi.org/10.1590/1807-1929/agriambi.v26n5p374-380

chan, f. k., & thong, j. y. (2009). acceptance of agile methodologies: a critical review and conceptual framework. *decision support systems*, *46*(4), 803-814.

fateh, m., djalal, m., & ammar, m. (2024). modeling and simulation of photovoltaic panel using simulink and proteus simulation. *journal of nano- and electronic physics*, *16*, 02028-02021. https://doi.org/10.21272/jnep.16(2).02028

gali, v. k., & solanki, s. (2025). leveraging hybrid agile methodologies for successful erp implementations. *international research journal of modernization in engineering technology and science*, *7*, 30. https://doi.org/10.56726/irjmets66269

johnpaul, s., selvan, c. t., & raguraman, p. (2022). iot based smart helmet system for accident prevention. 2022 international conference on edge computing and applications (icecaa),

kamal, r., david, l., & torshin, i. (2023). *embedded systems: architecture, programming, and design" by*. https://doi.org/10.13140/rg.2.2.20009.36961

khan, a. a. (2019). *smart helmet: automatic engine locking through alcohol detection* university of technology].

patel, p. (2021). *smart helmet for drunk and drive prevention with iot integration*.

romero, t. (2024). *lcd display i2c with python and embedded platforms*.

rubin, d., & perin, m. a. (2022). *buzzer alarm system with the help of arduino*.

sreeja, b., kumar, s. m., & sriram, s. (2021). smart helmet for alcohol detection to prevent from accidents. *annals of the romanian society for cell biology*, *25*(3), 2808-2815.

vijayan, s., govind, v. t., mathews, m., surendran, s., & sabah, m. e. (2014). alcohol detection using smart helmet system. *international journal of emerging technology in computer science & electronics*, *8*(1).

https://www.ijetcse.com/admin/uploads/alcohol%20detection%20using%20smart%20helmet%20system_1605764966.pdf

APPENDICES

APPENDIX A: ARDUINO IDE CODE

```
#include <TinyGPS++.h>

#include <EEPROM>

#include <SoftwareSerial.h>

#include <AltSoftSerial.h>

#include <Wire.h>

#include <LiquidCrystal_I2C.h>

SoftwareSerial gpsSerial(3, 4);

TinyGPSPlus gps;

AltSoftSerial gsmSerial;

LiquidCrystal_I2C lcd(0x27, 16, 2);

const int MQ3_PIN = A0;

const float ALCOHOL_THRESHOLD = 0.09;

const float ALCOHOL_BASELINE = 0.02;

const int NOISE_THRESHOLD = 5;

const int BUZZER_PIN = 2;

const int RED_LED_PIN = 10;

bool alertActive = false;

const String PHONE_NUMBER = "+250785866677";

const String VEHICLE_NAME = "Benitha RUTWE-X";

const String LICENSE_PLATE = "RAB 123A";

bool alertSent = false;


void setup() {
```

```
Serial.begin(9600);
while (!Serial);
gpsSerial.begin(9600);
gsmSerial.begin(9600);

lcd.init();
lcd.backlight();
lcd.print("Initializing...");

pinMode(BUZZER_PIN, OUTPUT);
pinMode(RED_LED_PIN, OUTPUT);
digitalWrite(BUZZER_PIN, LOW);
digitalWrite(RED_LED_PIN, LOW);

// Initialize GSM module
Serial.println("Initializing GSM module...");
if (!sendATCommand("AT", 2000)) {
Serial.println("GSM not responding!");
while(1);
}
sendATCommand("AT+CFUN=1", 1000);
sendATCommand("AT+CMEE=2", 1000);
sendATCommand("AT+CPIN?", 2000);
sendATCommand("AT+CSQ", 1000);
```

```
  sendATCommand("AT+CMGF=1", 1000);
  sendATCommand("AT+CNMI=1,2,0,0,0", 1000);

  lcd.clear();
  lcd.print("System Ready");
  Serial.println("System initialized");
}

void loop() {
  while (gpsSerial.available() > 0) {
    if (gps.encode(gpsSerial.read())) {
    if (gps.location.isValid()) {
        Serial.print("GPS Location: ");
        Serial.print(gps.location.lat(), 6);
        Serial.print(", ");
        Serial.println(gps.location.lng(), 6);
      }
    }
  }
  int rawValue = analogRead(MQ3_PIN);
  if (rawValue < NOISE_THRESHOLD) rawValue = 0;
  float alcoholLevel = rawValue * (5.0 / 1023.0) * 0.1 - ALCOHOL_BASELINE;
  if (alcoholLevel < 0) alcoholLevel = 0;
  lcd.clear();
```

```
lcd.setCursor(0,0);
lcd.print("BAC: ");
lcd.print(alcoholLevel, 3);
lcd.setCursor(0,1);
lcd.print(gps.location.isValid() ? "GPS: Locked" : "GPS: Searching");
if (alcoholLevel >= ALCOHOL_THRESHOLD) {
  if (!alertSent) {
    if (sendAlertSMS(alcoholLevel, gps.location.isValid())) {
      alertSent = true;
      lcd.clear();
      lcd.print("ALERT SENT!");
      Serial.println("Alert SMS sent successfully");
    } else {
      Serial.println("Failed to send alert!");
    }
  }
  alertActive = true;
  digitalWrite(RED_LED_PIN, HIGH);
  tone(BUZZER_PIN, 1000);
} else {
  alertActive = false;
  digitalWrite(RED_LED_PIN, LOW);
  noTone(BUZZER_PIN);
  alertSent = false;
```

```
  }

  delay(1000);
}

bool sendAlertSMS(float bac, bool gpsValid) {
  Serial.println("Preparing SMS...");
  String smsText = "DRUNK DRIVING ALERT\n";
  smsText += "Vehicle: " + VEHICLE_NAME + "\n";
  smsText += "Plate: " + LICENSE_PLATE + "\n";
  smsText += "BAC Level: " + String(bac, 3) + "\n";

  if (gpsValid) {
    smsText += "Location: " + String(gps.location.lat(), 6) + ", " +
String(gps.location.lng(), 6);
  } else {
    smsText += "Location: Not available";
  }

  gsmSerial.println("AT+CMGS=\"" + PHONE_NUMBER + "\"");
  if (!waitForPrompt(2000)) {
    Serial.println("Failed to get SMS prompt");
    return false;
  }
```

```
  gsmSerial.print(smsText);
  delay(100);
  gsmSerial.write(26); // Ctrl+Z
  return waitForConfirmation(10000);
}

bool sendATCommand(String cmd, unsigned long timeout) {
  Serial.println("CMD: " + cmd);
  gsmSerial.println(cmd);
  return waitForResponse(timeout);
}

bool waitForResponse(unsigned long timeout) {
  unsigned long start = millis();
  String response;
  while (millis() - start < timeout) {
    while (gsmSerial.available()) {
    char c = gsmSerial.read();
    Serial.write(c);
      response += c;
      if (response.indexOf("OK") != -1) return true;
      if (response.indexOf("ERROR") != -1) return false;
    }
```

```
  }
  return false;
}
bool waitForPrompt(unsigned long timeout) {
  unsigned long start = millis();
  String response;
  while (millis() - start < timeout) {
    while (gsmSerial.available()) {
    char c = gsmSerial.read();
    Serial.write(c);
      response += c;
      if (response.indexOf("> ") != -1) return true;
      if (response.indexOf("ERROR") != -1) return false;
    }
  }
  return false;
}
bool waitForConfirmation(unsigned long timeout) {
  unsigned long start = millis();
  String response;
  while (millis() - start < timeout) {
    while (gsmSerial.available()) {
    char c = gsmSerial.read();
    Serial.write(c);
```

```
      response += c;
      if (response.indexOf("+CMGS") != -1) return true;
      if (response.indexOf("ERROR") != -1) return false;
    }
  }
  return false;
}
```